OFFICIAL OPERATIONS MANUAL • ENTERPRISE EDITION

NETSPECTRE

Visual Network Intelligence, Multi-Protocol Diagnostics & Endpoint Troubleshooting Platform

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This document provides comprehensive operational specifications for NETSPECTRE V3.0 Enterprise, detailing the Tri-Persona Workspace system (Personal, Developer, Network Operations), native Win32 kernel telemetry architecture, live REST APIs, interactive WebGL neural flow visualization, and deep troubleshooting suites.

APPLICATION VERSION	TARGET PLATFORM	TELEMETRY ENGINE	CLASSIFICATION
v3.0 ENTERPRISE	Windows 10 / 11 x64	Win32 IP Helper	Production Grade

1. EXECUTIVE OVERVIEW & PERSONA WORKSPACES

NETSPECTRE V3.0 is an enterprise-grade Windows desktop application that bridges the gap between low-level system packet dynamics and user-facing actionable intelligence. Operating on a **100% genuine Win32 telemetry foundation**, NETSPECTRE queries the Windows IP Helper API (`iphlpapi.dll`), kernel process handles, and network adapter hardware counters.

The Tri-Persona Workspace Architecture

NETSPECTRE V3.0 eliminates interface clutter by tailoring tools and views to three distinct user personas via the top-bar workspace switcher:

👤 Personal Workspace

Everyday Windows Users & Remote Workers: 1-Click Network Health Doctor (0-100 Quality Index), plain-English explanations ("Why is my internet slow?"), bandwidth consumption meters, background process trackers, and 1-click Windows Defender Firewall application blocking.

Developer Workspace

Software & API Engineers: Interactive REST API Debugger with microsecond phase waterfalls (DNS $\rightarrow$ TCP $\rightarrow$ TLS $\rightarrow$ TTFB $\rightarrow$ Download), automated secret masking, cURL/HAR exports, Failure Root-Cause Analyzer, and multi-stage Environment Health Monitoring (DEV/SIT/UAT/PROD).

🌐 Network Operations Workspace

IT Administrators, Network Engineers & SOC Analysts: 2.5D Orthographic Neural Highway with 5 specialized visual modes, interactive node right-click context menus, deep Troubleshooting Workbench (Ping, Traceroute, DNS, Port Prober, TLS Inspector, Routing Table, Adapters), Incident Forensics Recorder, Multi-Device Monitor, and binary PCAP generation for Wireshark.

Core Operating Principles

- **100% Genuine Win32 Telemetry:** Every displayed socket, PID, protocol, and throughput metric reflects actual kernel data. No simulated or fictional connections.
- **Cryptographic Truthfulness:** HTTPS and QUIC flows are encrypted via TLS 1.3. NETSPECTRE measures connection metadata, flow rates, and endpoints accurately without claiming to decrypt private payload contents.
- **100% Local-First Security:** Zero cloud telemetry exfiltration. All measurements, logs, and captures remain local on the user's workstation.

2. PERSONAL WORKSPACE — NETWORK DOCTOR

The **Network Doctor** (`network-doctor`) provides a one-click automated health checkup of your workstation's complete network path. It benchmarks local router reachability, public internet backbone latency, DNS resolution speed, and physical Wi-Fi signal quality.

Health Score & Quality Index (0–100)

The Network Doctor computes a holistic 0–100 score updated dynamically:

SCORE RANGE	STATUS RATING	COLOR INDICATOR	OPERATIONAL MEANING
85 – 100	EXCELLENT	Emerald Green	Optimal link quality. Latency < 50ms, DNS < 60ms, Wi-Fi > 75%. Ready for competitive gaming & 4K streaming.
65 – 84	GOOD	Cyan Sky	Normal web browsing and video conferencing. Moderate latency without noticeable dropped packets.
40 – 64	DEGRADED	Amber Orange	High latency or weak Wi-Fi signal. Video buffering and sluggish API calls expected.
0 – 39	CRITICAL	Rose Red	Gateway unreachable, internet offline, or DNS lookup failure. Network intervention required.

The Four Pillars of Diagnostic Health

1. Local Router / Default Gateway

Probes the physical gateway IP (e.g., `192.168.1.1` or `10.70.107.182`). Detects router freezes, physical cable disconnection, or local link drops.

2. Global Internet Backbone

Measures ICMP round-trip latency against Tier-1 anycast backbones (`1.1.1.1` and `8.8.8.8`) to evaluate ISP upstream health.

3. DNS Name Resolution

Measures domain lookup duration against `cloudflare.com` to verify fast domain-to-IP translation.

4. Physical Link & Wi-Fi Quality

Extracts signal percentage, SSID, and 802.11 radio protocol via native `netsh wlan show interfaces`.

> **Actionable Findings:** Each diagnostic check includes plain-English explanations and 1-click remediation buttons that launch the Troubleshooting Workbench with the failing target pre-filled.

3. INTERNET USAGE, PRIVACY & FIREWALL CONTROLS

Internet Usage & Privacy Monitor

The **Internet Usage Monitor** (`internet-usage`) provides full transparency into per-application bandwidth consumption and background privacy exposures:

- **Bandwidth Share Progress Bars:** Visual breakdown of each application's cumulative byte volume as a percentage of total session data.
- **Background Data Consumer Identification:** Flags background applications moving significant traffic without open UI windows.
- **Tracker & Telemetry Audit:** Scans active socket domains for known analytics and advertising trackers (e.g. telemetry endpoints, metrics, doubleclick), allowing immediate review.

Application Internet Firewall Controls

The **Firewall Manager** (`firewall`) allows users to restrict outbound network access for any Windows executable with a single click:

```
netsh advfirewall firewall add rule name="NETSPECTRE Block - [App]" dir=out action=block program="[Path]"
enable=yes
```

Native Defender Integration

Rules are written directly to the Windows Defender Firewall engine. Rules persist even when NETSPECTRE is closed.

Security Audit Log

Maintains an in-memory chronological record of every block and unblock action with timestamps and process paths.

ACTION	TARGET PROCESS	FIREWALL RULE CREATED	STATUS
BLOCK	telemetry_daemon.exe	NETSPECTRE Block - telemetry_daemon.exe	Outbound Traffic Blocked
BLOCK	untrusted_agent.exe	NETSPECTRE Block - untrusted_agent.exe	Outbound Traffic Blocked
UNBLOCK	chrome.exe	[Deleted from Windows Defender Rules]	Outbound Traffic Restored

4. DEVELOPER WORKSPACE — API TRAFFIC DEBUGGER

The **API Traffic Debugger** (`api-debugger`) empowers software developers to execute HTTP/HTTPS requests with microsecond phase waterfall timing, response body inspection, and automated credential protection.

Microsecond Phase Waterfall Breakdown

Every executed request is profiled across 5 discrete timing phases:

PHASE	COLOR	WHAT IT MEASURES	COMMON BOTTLENECKS
DNS Lookup	Blue	Time required to resolve domain name to IP address.	Slow local resolver, DNS server congestion.
TCP Handshake	Emerald	Time for SYN, SYN-ACK, ACK 3-way connection setup.	Physical distance, high routing latency.
TLS Handshake	Purple	SSL/TLS certificate negotiation and cipher exchange.	Slow crypto negotiations, deep certificate chains.
TTFB (Wait)	Amber	Time To First Byte while server computes response.	Slow database queries, heavy backend microservices.
Content Download	Cyan	Time to transfer the response payload stream.	Large JSON payloads, low client downlink bandwidth.

Developer Tooling Integration

- **1-Click Copy as cURL:** Generates ready-to-use cURL commands formatted with headers, HTTP method, and JSON body for terminal replay.
- **Export HAR (HTTP Archive):** Generates standard HAR 1.2 files compatible with Chrome DevTools, Postman, and Insomnia.
- **Automated Secret Masking:** Sensitive authorization headers (`Bearer`, `Authorization`, `Cookie`, API keys) are sanitized automatically in stored execution history.

5. API FAILURE ANALYZER & ENVIRONMENT MONITOR

API Failure Root-Cause Analyzer

The **API Failure Analyzer** (`api-failure-analyzer`) provides deterministic automated diagnosis for failed requests:

DNS_RESOLUTION_FAILURE (NXDOMAIN)

Domain name does not exist or local resolver failed. Verified via Workbench DNS tool. Suggests `ipconfig /flushdns`.

TCP_CONNECTION_REFUSED (ECONNREFUSED)

Target IP is alive but no service is listening on port. Suggests checking port binding and container status.

TLS_CERTIFICATE_EXPIRED

Server certificate expired or hostname mismatch. Verified via Workbench TLS Certificate Inspector.

HTTP 502 / 504 GATEWAY TIMEOUT

Reverse proxy (Nginx, ALB, Cloudflare) timed out waiting for backend microservice. Suggests inspecting container OOM logs.

Multi-Stage Environment Health Monitor

The **Environment Monitor** (`env-monitor`) tracks application endpoints across DEV, SIT, UAT, and PROD:

TIER	SERVICE NAME	TARGET URL	LATENCY	SSL EXPIRY	STATUS
PROD	Production API Gateway	api.github.com	24 ms	184 days remaining	ONLINE
UAT	Staging Cloudflare Edge	cloudflare.com	16 ms	98 days remaining	ONLINE
SIT	Integration Testing Hub	httpbin.org	62 ms	240 days remaining	ONLINE
DEV	Local Development Host	127.0.0.1:48123	1 ms	365 days remaining	ONLINE

6. OPERATIONS WORKSPACE — 2.5D NEURAL HIGHWAY

The **Neural Highway Universe** (`command-center`) transforms network traffic into an interactive 2.5D orthographic graph.

The 5-Column Information Architecture

COLUMN	LAYER NAME	ENTITIES DISPLAYED	VISUAL ATTRIBUTES
Column 1	LOCAL SYSTEM	Local PC, Active Network Interface (Wi-Fi, Ethernet).	Cyan pulsing ring with live uplink/downlink gauges.
Column 2	APPLICATIONS	Detected running Win32 processes (Chrome, Edge, Teams, VS Code, etc.).	Sized by active socket count and transfer volume.
Column 3	PROTOCOLS & HUBS	Transport protocols (TCP, UDP) and port aggregators.	Cyan disc clusters showing active socket bindings.
Column 4	DESTINATIONS	Remote IPs, domain names, country flags, organizations.	Pulsing destination discs with throughput counters.
Column 5	SERVICE / ASN	Autonomous System Numbers (Cloudflare, AWS, Google, Microsoft).	Purple routing cluster nodes showing BGP networks.

Interactive Right-Click Context Menu

Right-clicking any node on the 2.5D WebGL canvas opens an instant context menu:

- **Destination Nodes:** 1-click jump to Workbench Ping, Traceroute, DNS Inspector, Port Prober, or TLS Inspector with target pre-filled.
- **Application Nodes:** 1-click "Block in Windows Firewall" or "Inspect in Application Explorer".
- **Universal:** "Diagnose with Network Doctor" for automated health verification.

7. NETWORK TROUBLESHOOTING WORKBENCH

The **Network Workbench** (workbench) provides a unified interface for 7 essential network troubleshooting utilities:

1. Ping Test Tool

Real Windows ICMP pings. Measures sequence numbers, RTT (ms), TTL, packet loss percentage, jitter, and min/avg/max latencies.

2. Traceroute Tool

Hop-by-hop ICMP TTL progression from 1 to 20 hops. Resolves intermediate router IPs and reverse hostnames with latency breakdown.

3. DNS Inspector

Queries A, AAAA, and CNAME records with sub-millisecond query timers using system and configured DNS resolvers.

4. TCP Port Prober

Tests 3-way handshake connectivity to ports (80, 443, 22, 3306, 5432, 8080) and reads service greeting banners.

5. TLS Certificate Inspector

Completes client TLS handshake with SNI. Extracts Subject, Issuer, Validity Dates, Days Until Expiry, Cipher Suite, and SANs.

6. System Routing Table

Parses the Windows IPv4 Routing Table (Destination, Netmask, Gateway, Interface, Metric) for diagnosing routing loops.

7. Physical & Virtual Adapter Inspector

Displays all detected adapters (Ethernet, Wi-Fi, VPN), operational status, MAC address, link speed (Mbps), MTU, and IP assignments.

8. INCIDENT RECORDER, INFRASTRUCTURE & PCAP

Incident Recorder & Forensic Snapshot Audit

When a network outage, VPN drop, or latency burst occurs, click **"CAPTURE INCIDENT SNAPSHOT"** in the Incident Recorder (`incident-recorder`):

- Freezes all active TCP/UDP connections, process bandwidth consumption, and interface counters at that microsecond.
- Generates an exportable post-mortem report in **Markdown** or **JSON** format for team retrospectives and ticket attachments.

Infrastructure & Multi-Device Monitoring

The **Infrastructure Monitor** (`devices`) continuously polls critical nodes:

DEVICE	TARGET IP / HOST	ROLE	STATUS
Default Local Gateway	192.168.1.1 / 10.70.107.182	Local Router	ONLINE (0 ms)
Cloudflare Anycast DNS	1.1.1.1	Primary Resolver	ONLINE (15 ms)
Google Public DNS	8.8.8.8	Secondary Resolver	ONLINE (16 ms)
GitHub Global CDN	140.82.112.4	Git Infrastructure	ONLINE (24 ms)
AWS Core Infrastructure	52.95.110.1	Cloud Services	ONLINE (28 ms)

Binary PCAP Generation

NETSPECTRE maintains a circular packet buffer and exports standard **binary PCAP capture files** (`netspectre-capture.pcap`) via `GET /api/pcap/export` for deep packet inspection in **Wireshark**.

9. REST API REFERENCE & OPERATIONS FAQ

METHOD	ENDPOINT PATH	DESCRIPTION
GET	/api/telemetry	Full JSON snapshot (connections, processes, rates, events).
GET	/api/diagnose/doctor	Network Doctor diagnostic results (Gateway, Internet, DNS, Wi-Fi, Score).
GET	/api/workbench/ping?target=...&count=4	ICMP Ping results with min/avg/max/jitter statistics.
GET	/api/workbench/traceroute?target=...	ICMP TTL-increment traceroute with reverse hostnames.
GET	/api/workbench/dns?host=...&type=A	DNS query records (A, AAAA, CNAME) and duration.
GET	/api/workbench/port?host=...&port=...	TCP 3-way handshake connect latency and service banner.
GET	/api/workbench/tls?host=...&port=443	TLS certificate validity, expiry days, SANs, and cipher suite.
POST	/api/firewall/block	Adds outbound executable block rule via netsh advfirewall.
POST	/api/api-inspector/execute	Executes HTTP/HTTPS request with phase waterfall metrics.
POST	/api/incident/snapshot	Freezes current state and compiles forensic incident report.

Frequently Asked Questions

Q: Why does the Network Doctor report 0ms gateway latency?
A: On fast gigabit Ethernet or local router switches, round-trip echo latency is sub-millisecond, reported as 0ms.

Q: Do firewall rules stay active after closing NETSPECTRE?
A: Yes. Rules are written directly to Windows Defender Firewall. They can be unblocked anytime from NETSPECTRE or Windows Control Panel.

Q: Can NETSPECTRE decrypt HTTPS bank traffic?
A: No. NETSPECTRE preserves cryptographic integrity. Modern TLS 1.3 encryption protects payloads; NETSPECTRE monitors socket flow metrics, metadata, and endpoints truthfully.